

Welcome to A Catholic Introduction to Artificial Intelligence Class 8! AI in Gaming & Entertainment

1. Overview

This class covers AI's role in gaming (e.g., NPC behavior, procedural content generation) and entertainment (AI-generated music, deepfake videos). Students will explore how AI tools are used to create digital art and animation. A hands-on activity will allow them to generate AI-created music or images. We will discuss whether AI-created content can be considered "real" creativity.

2. Learning Objectives

By the end of this class, students should be able to:

1. Explain how AI is used in modern video games
2. Describe procedural content generation and NPC behavior
3. Identify examples of AI in music, art, film, and animation
4. Explain what deepfakes are and how they are created
5. Generate simple AI-created media using online tools
6. Discuss whether AI-generated media qualifies as creativity

3. Class 7 Review

In Class 7, we explored how AI is used in healthcare and medicine. We learned that AI systems can help doctors analyze medical images such as X-rays, MRIs, and CT scans more quickly and accurately. AI can identify patterns in large amounts of medical data that might be difficult for humans to notice. Hospitals also use AI to assist with scheduling, predicting patient risks, and helping researchers discover new medicines. While AI can be a powerful medical tool, we discussed that human doctors still make the final decisions and provide the human judgment and compassion that machines cannot replace.

We also discussed how modern AI systems learn from data. Many healthcare AI systems are trained using thousands or even millions of medical examples. This allows them to recognize patterns and improve over time. We reviewed how large language models and other AI systems are built from data collected from books, websites, articles, images, and other digital information. Students learned that the quality and accuracy of data are important because AI systems can only learn from the information they are given.

Finally, we examined ethical and moral questions related to AI in healthcare. We discussed concerns about privacy, bias, and trust. For example, if an AI system is trained using incomplete or unfair data, it may produce inaccurate or biased results. We also considered how Catholics and society in general should think carefully about the proper role of technology in human life. AI can assist people in important ways, but it should always serve human dignity and the common good rather than replace human responsibility or moral judgment.

4. AI in Video Games

Video games began in the 1970s with simple arcade games such as Pong and later home console systems like the Atari 2600. Early games were limited by the technology of the time and used very simple rules and graphics. Enemies often followed predictable patterns rather than acting intelligently. As computers became more powerful in the 1980s and 1990s, game developers began using more advanced programming techniques to make games feel more realistic and challenging. Games such as Pac-Man used basic `AI` to control enemy movement, while strategy and role-playing games introduced more sophisticated computer opponents.

As gaming technology improved, artificial intelligence became an increasingly important part of game design. Developers used `AI` to control non-player characters (NPCs), create realistic enemy behavior, and generate dynamic game worlds. Procedural content generation allowed games like Minecraft and No Man's Sky to create enormous worlds automatically using algorithms and randomness. Modern games also use AI for pathfinding, adaptive difficulty, matchmaking, and anti-cheat systems. Some games even study player behavior to adjust gameplay and improve engagement.

Interestingly, when gaming refers to itself using `AI` it is referring to making characters appear `intelligent` and not using and specific academic `AI`.

Today, AI is expanding beyond gameplay itself and into the creation of games. Developers now use generative AI tools to create artwork, music, dialogue, animations, and voices; this is the real academic AI being used.

Let's take a look at `NPC behavior`, `procedural content generation`, `adaptive game play`, `AI-Assisted Game Development`, and `AI as the "Director"`.

4.1 NPC Behavior

Non-player characters (NPC) as well as all gaming characters for that matter use `finite state machines` (FSM) and `rules-based algorithms` to make them appear intelligent.

Simple rule-based FSMs work well for predictable NPCs. Modern games increasingly add more sophistication:

- Probabilistic behavior — NPCs have a chance of choosing between several actions, making them feel less robotic
- Behavior trees — a structured hierarchy of decisions that allows richer, more nuanced NPC logic
- Machine learning — in some games, NPCs are trained on player data so they can adapt over time

For most games you have played, the NPC behavior was driven by hand-crafted rules and probabilities, not machine learning. But that boundary is shifting as development tools improve.

4.2 Procedural Content Generation

Procedural generation means using algorithms — rather than human designers — to automatically create game content. Instead of an artist hand-crafting every room, every weapon, or every landscape, a set of mathematical rules produces the content automatically, often using a seed value as a starting point.

The practical benefit is enormous: a game can contain far more content than any team of designers could create manually. In some cases, the content is effectively infinite.

Procedural techniques can be applied to almost any type of game content:

- Maps and levels — dungeon layouts, room arrangements, and pathways generated fresh for each playthrough
- Terrain — landscapes, mountains, rivers, caves, and biomes produced by mathematical noise functions
- Infinite worlds — content that generates on demand as the player explores, rather than all at once

- Loot and items — weapon properties, item stats, and drop rates calculated on the fly
- Quests and events — some games generate mission objectives and story events procedurally

Minecraft is perhaps the most widely known example of procedural world generation. Every world is produced from a single seed number. That seed is fed into a noise function — a mathematical formula producing smooth, natural-looking variation — which determines terrain height, biome placement, cave locations, and resource distribution. The result is a world that is different every playthrough (unless you use the same seed), effectively infinite, internally consistent, and surprisingly natural-looking. No human designer created any individual chunk of that world.

4.3 Adaptive Game Play

Modern games do not simply respond to your inputs. Many actively study you — collecting data about how you play and using that data to shape your experience in real time. This is AI functioning as an analytics and prediction system, similar in principle to the recommendation engines used by streaming services and online retailers.

The types of data a game might track include:

- Player skill — accuracy, reaction time, win rate, and completion rate
- Decision patterns — which routes, weapons, or strategies you prefer
- Frustration signals — repeated deaths, long pauses, quitting and returning
- Engagement time — how long you play per session and how often you return
- Spending behavior — in free-to-play games, what you purchase and when

One of the most common forms of adaptive gameplay is Dynamic Difficulty Adjustment (DDA). Rather than offering fixed settings like Easy, Normal, and Hard, DDA continuously recalibrates challenge based on actual performance. In practice:

- If you die repeatedly, enemies become slightly weaker, drop more health, or become less accurate
- If you are dominating, enemies become faster, more aggressive, or more numerous
- These adjustments are typically subtle and invisible — most players do not notice them

In free-to-play games with in-app purchases, adaptive AI sometimes serves a more commercially motivated role. The AI analyses spending behavior and engagement patterns, then adjusts the experience to maximize the likelihood of a purchase. For example:

- Increasing difficulty just before a point where powerful items are sold
- Sending notifications when engagement is predicted to be dropping
- Offering personalized deals on items the player has looked at but not purchased

This raises important questions. The same techniques that make a game more enjoyable can also be used to exploit psychological vulnerabilities — particularly in younger players.

Catholic social teaching reminds us that persons should never be treated as data points or means to profit, but always as ends in themselves with inherent dignity.

4.4 AI-Assisted Game Development

The previous sections described AI systems built into games, which players encounter directly. This section is different. It describes how AI tools are changing the process of making games — and it is arguably the most significant development in the games industry right now. Game development has traditionally required large, multidisciplinary teams working for years. Generative AI is beginning to compress that timeline, allowing smaller teams — or even individuals — to produce content at a scale that would previously have required dozens of specialists.

Studios are now using AI tools to assist with:

- Art assets and textures — concept art, environment textures, character designs, and UI graphics
- Dialogue and narrative — NPC conversations, quest text, lore documents, and branching storylines
- Voice acting — AI-generated or AI-cloned voices for characters
- Music and sound — dynamic background scores and adaptive soundtracks
- Animation — character movement and facial expressions generated from text descriptions
- Code — game logic, tools, shaders, and entire systems generated from natural-language prompts

Some developers can now prototype entire games with generative AI assistance in hours rather than months. This is not a distant future scenario — it is happening in studios of all sizes today. Independent developers who previously could not afford original art or music can produce polished-looking games with minimal budget.

4.5 AI as the "Director"

Everything above described AI at the character level: a single NPC making decisions about what to do. The AI Director is a fundamentally different concept. It operates above all individual characters, watching the entire game session like a film director watching a production — and quietly shaping what happens next.

The goal is to produce an experience that feels emotionally well-paced: tension builds, reaches a peak, releases, and then builds again. This structure underlies almost all compelling drama. Achieving it in a game is difficult, because players can do unpredictable things.

An AI Director typically has influence over:

- Enemy spawning — when enemies appear, in what numbers, and how aggressively they behave
- Resource placement — where health packs, ammunition, and power-ups appear
- Pacing — the overall rhythm of intensity and relief across the session
- Atmospheric cues — music, lighting, and sound that signal danger or safety

Left 4 Dead (Valve, 2008) contains one of the best-known examples of an AI Director in a commercial game. The Director monitored several signals in real time: how much damage each player had taken, how many resources players had available, how long since the last major encounter, and whether the group was moving quickly or becoming stuck.

Based on this data, the Director adjusted zombie attack frequency and intensity, the placement of health kits and weapons, and the timing of special enemy appearances. If a group was struggling, the Director reduced pressure. If they moved through a level too easily, it escalated. The result: every session felt different and appropriately tense, even on a map players knew well. The Director was not scripted to trigger specific events at specific moments; it was responding dynamically to each group's actual experience.

A guard NPC decides what to do within a local situation. It sees the player and decides to chase. Its scope is narrow and its decision is immediate. The AI Director watches the entire session. It has no local situation — it sees everything at once and operates on a timescale of minutes and hours, not seconds. Its goal is not to win an encounter but to ensure the player has a satisfying emotional experience. This distinction matters. Simple NPC behavior is well understood and widely used. True AI direction — monitoring the whole experience and adjusting it dynamically — is a significantly more sophisticated problem.

4.6 Vector Math in Gaming and in Machine Learning

Perhaps the most interesting connection between gaming and AI is that decades-old mathematical algorithms now sit at the core of machine learning. Both gaming and machine learning use vectors to represent position, direction, relationships, and transformations in multi-dimensional space. The difference is mostly what the vectors represent.

Vector Math in Gaming

In games, vectors usually represent physical quantities in 2D or 3D space. A game character's position is described by a coordinate triplet (x, y, z) — where x is left/right, y is up/down, and z is forward/backward. If an enemy walks toward the player, the game computes a direction vector describing the path between them. Games constantly perform vector operations to power their core systems:

- Normalization — scaling a vector to unit length to represent pure direction
- Dot products — determining angles, lighting intensity, and line-of-sight
- Cross products — computing normals for lighting and surface rendering
- Distance calculations — collision detection, proximity checks, and pathfinding

Together these operations power aiming, physics, collision detection, lighting, animation, and camera movement — the basic machinery of almost every 3D game.

Vector Math in Machine Learning

In AI training, vectors represent features or information rather than physical space. A word embedding vector might look like (0.12, −0.44, 1.83, 0.91, ...) — but these numbers do not represent a position in a game world. Instead, they encode meaning, relationships, patterns, and probabilities that the model has learned from its training data.

In machine learning, almost everything becomes a vector. Images become vectors of pixel intensities. Words become vectors in a high-dimensional semantic space. Sounds become vectors of frequency data. Even individual users become vectors representing their preferences and behavior patterns. The AI learns relationships between those vectors — and those learned relationships are what we call "understanding."

The Key Similarity: Distance and Direction

This is the deep connection between the two fields: both systems operate in mathematical spaces where distance and direction carry meaning. In a game, distance between two vectors may mean physical closeness — two enemies near each other in the level. In an AI model, distance between two vectors may mean conceptual similarity — two words that are related in meaning.

A concrete example: in a game, two nearby enemies are physically close in 3D space. In a large language model, the words "king" and "queen" are mathematically close in semantic space — their vectors point in similar directions. The mathematics is identical; only what the space represents differs. That is a powerful conceptual bridge for students.

Dot Products: A Shared Tool

One of the most striking overlaps between game math and AI math is the dot product. The formula is:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(\theta)$$

In a game engine, the dot product determines the angle between two vectors. This is used to calculate lighting intensity (how directly does light hit a surface?), whether an enemy can see the player (is the player within the guard's field of view?), and physics interactions. In machine learning, the dot product measures similarity between vectors — how related are two words, how close are two data points? Attention mechanisms in transformer models — the architecture behind large language models — rely heavily on dot products to determine which parts of the input the model should focus on. Transformers and game engines are both, at their core, executing enormous numbers of dot products.

GPUs: The Hardware Connection

This is probably the most important practical connection between gaming and AI. Gaming drove the development of graphics processing units (GPUs) because rendering 3D graphics requires massive quantities of vector and matrix calculations performed in parallel — thousands of pixels being computed simultaneously, every frame, at high speed.

When AI researchers began scaling up neural networks, they realized that training a neural network requires exactly the same kind of math: enormous numbers of vector and matrix operations, all of which benefit from parallel computation. The chips that decades of gaming had pushed NVIDIA and AMD to develop turned out to be ideal for AI training. That is why NVIDIA became one of the most valuable companies in the world almost overnight when large language models took off. The gaming industry had, without knowing it, built the hardware infrastructure that AI needed.

The underlying mathematical overlap is significant: both game rendering and neural network training rely on vectors, matrices, parallel computation, and linear algebra. The only difference is what those operations are computing.

Even in Minecraft, vectors are everywhere — controlling player movement, camera direction, physics, and mob pathfinding. This makes Minecraft a useful bridge. When discussing the game, point out that every action the player takes is being translated into vector operations by the engine. Then make the leap: an AI model generating Minecraft-like worlds uses vectors too, but those vectors represent terrain patterns, texture relationships, structural conventions, and player behavior tendencies. The same mathematical tools appear in both contexts, applied to fundamentally different kinds of information.

5. AI in Entertainment and Media

This section examines how artificial intelligence has become deeply embedded in the entertainment and media industries. AI now shapes what music you hear, what films you watch, what news you read, and what content appears on your social media feed. It can generate images and music indistinguishable from human-made work, clone voices with seconds of audio, and fabricate video footage of people saying things they never said.

These capabilities offer genuine benefits — more personalized experiences, lower barriers to creative production, more efficient journalism (but perhaps not more *truthful*). They also introduce serious risks: the erosion of trust in what we see and hear, the manipulation of public opinion, the displacement of human creativity, and the concentration of extraordinary power in a small number of technology platforms.

5.1 Streaming Recommendations

When you open Netflix, Spotify, or YouTube, the first thing you see is not a neutral catalogue. It is a personalized list, assembled by an AI system that has been predicting what

you are most likely to watch, listen to, or click on next. This prediction is based on enormous amounts of data about you and about millions of other users.

The core technique is called collaborative filtering. The system looks at your history — what you have watched, how long you watched it, what you skipped, what you returned to — and finds other users whose patterns closely resemble yours. It then recommends content that those similar users enjoyed but that you have not yet seen. The underlying assumption is that people with similar tastes in the past will have similar tastes in the future.

It is important to understand what recommendation systems are designed to maximize. In most commercial contexts, the goal is engagement: the number of hours you spend on the platform, the number of items you consume, the likelihood that you will subscribe or remain subscribed.

Engagement and quality are not the same thing. Content that provokes strong emotional reactions — outrage, fear, excitement, lust — tends to generate more engagement than content that is informative, nuanced, or morally serious. A recommendation system optimizing for engagement may therefore systematically surface content that is emotionally provocative rather than genuinely valuable. This matters particularly for younger users, whose habits, tastes, and worldviews are still being formed. The content a platform recommends is not neutral. It shapes what feels normal, what feels desirable, and what feels true.

The Filter Bubble Problem A further concern is what researchers call the filter bubble. When a recommendation system consistently surfaces content that aligns with your existing preferences and beliefs, it narrows your exposure to different perspectives. Over time, you may find yourself in a media environment that confirms everything you already think and challenges nothing - kind of like an *echo-chamber* of viewing.

This is not necessarily the intent of the algorithm's designers. It is an emergent consequence of optimizing for engagement: familiar, agreeable content produces more clicks than challenging, uncomfortable content. But the effect on how we understand the world — and on our capacity for genuine dialogue with people who see things differently — can be significant.

5.2 AI Music

Generative AI models trained on large datasets of human-created music, images, text, and video can now produce new creative work that is, in many cases, indistinguishable from human output. These systems do not understand what they are creating in any meaningful sense. They identify statistical patterns in their training data and generate new outputs that follow those patterns. The capabilities are advancing rapidly:

- AI music tools can compose original pieces in any genre, generate backing tracks, continue an existing melody, or clone the style of a specific artist

- AI image generators can produce photorealistic images, illustrations, paintings, or designs from text descriptions in seconds
- AI video tools can generate short video clips, animate still images, or apply consistent visual styles across footage
- AI voice tools can clone a person's voice from a few seconds of audio, producing new speech in their voice that they never recorded

These tools lower the barriers to creative production significantly. A student with no formal music training can produce a polished backing track. An independent filmmaker can create visual effects that previously required a studio budget. A small organization can produce professional visual content without hiring designers.

There are also genuinely new creative possibilities. AI can generate variations on a theme faster than any human, explore stylistic combinations that would not occur to a human creator, and assist artists in realizing visions they could not execute technically on their own.

These capabilities raise profound questions about creativity, authorship, and economic justice. Most generative AI models are trained on vast datasets of human-created work — music, art, writing, photography — often without the knowledge or consent of the creators and without any compensation. Artists, musicians, and writers whose work trained these systems find their livelihoods threatened by tools built, in part, on their own labour.

There are also questions of attribution and deception. When an AI generates an image in the style of a living artist, is that fair use or plagiarism? When a streaming platform fills playlists with AI-generated music to avoid paying royalties to human musicians, who benefits and who is harmed? When AI-generated art wins competitions or AI-written articles are published without disclosure, what happens to the social value we attach to human creative achievement?

5.3 Deepfakes

A deepfake is a piece of audio, video, or image content in which a real person's face, voice, or likeness has been replaced or fabricated using AI. The term combines deep learning — the AI technique involved — with fake. The results can be extraordinarily convincing. The underlying technology uses a class of AI models called generative adversarial networks (GANs). Two neural networks work against each other: one generates fake content, the other tries to detect whether it is fake. Over many cycles, the generator learns to produce output convincing enough to fool the detector — and, eventually, convincing enough to fool human observers. Current deepfake capabilities include:

- Face replacement in video — placing one person's face onto another person's body with realistic lighting, movement, and expression
- Voice cloning — generating speech in a specific person's voice from a short audio sample
- Lip synchronization — matching mouth movements to fabricated speech

- Fully synthetic faces — generating photorealistic faces of people who do not exist
- Text-to-video — generating video footage of a person saying or doing something from a text description

Deepfakes present risks across several distinct domains.

Political disinformation

Fabricated video of a political leader making statements they never made, released at a critical moment, could influence elections, incite conflict, or undermine legitimate institutions. The risk is not only that people believe false content — it is that the existence of deepfakes gives people reason to doubt genuine content.

Non-consensual intimate imagery

The majority of deepfake content found online is non-consensual sexual material, in which a real person's face is placed on explicit content without their knowledge or consent. This is a serious violation of human dignity and causes profound harm to victims, predominantly women.

Financial fraud

Voice cloning has been used to impersonate executives and instruct employees to transfer funds. In several documented cases, organisations lost significant sums to audio deepfakes of their own leadership.

Erosion of trust

Perhaps the most insidious long-term risk is not any specific piece of fake content, but the general erosion of trust in audiovisual evidence. When people can no longer rely on video as proof of what happened, the shared factual foundation of public life is weakened.

Detection and Response

AI detection tools exist and are improving. Researchers look for artifacts in deepfake video — subtle inconsistencies in lighting, blinking patterns, or facial geometry that are difficult to eliminate entirely. Some platforms require disclosure of AI-generated content. Several countries have introduced or are considering legislation.

However, detection tools consistently lag behind generation capabilities. As generation improves, detection must keep pace. This is, fundamentally, an arms race — and one with serious implications for public trust in media.

https://thispersondoesnotexist.com/

5.4 Hollywood

AI has become a significant tool in film and television production, primarily in post-production — the editing, effects, and finishing work done after filming is complete. Some of the most visible applications involve manipulating the appearance of actors. De-ageing technology, used prominently in recent films, uses AI to smooth an actor's features to match their younger appearance. The same techniques can be used to age actors, alter their physical appearance, or in some cases recreate the likeness of deceased performers. AI is also used in:

- Visual effects and compositing — separating subjects from backgrounds, generating realistic environments, and creating creatures or objects that do not exist
- Color grading — automatically adjusting the color and tone of footage to match a desired look
- Noise reduction and restoration — improving the quality of footage shot in difficult conditions, or restoring damaged archival film
- Dubbing and lip synchronization — adapting dialogue for international audiences while matching mouth movements

Less visible but equally significant is the use of AI in creative and business decisions. Studios use AI tools to analyze scripts, predict box office performance, identify which story elements have historically performed well with specific audiences, and recommend casting choices based on data. This introduces a tension that is worth examining carefully. Film and television at their best tell stories that are surprising, challenging, and morally serious — stories that audiences did not know they needed until they encountered them. An AI optimizing for predicted commercial performance tends to recommend the familiar, the proven, and the safe. It cannot easily predict the impact of genuinely new creative work, because genuinely new work has no precedent in the training data.

In 2023, writers and actors in Hollywood went on strike in part over the use of AI in production. Among the specific concerns were: studios using AI to generate first drafts of scripts that human writers would then revise for minimum pay, AI scanning an actor's likeness to generate footage without additional compensation, and the use of AI to perform background roles that would previously have employed human performers.

These concerns reflect a broader pattern visible across many industries: AI enabling the extraction of value from human creative work while reducing the need to compensate the human beings who made that work possible.

MASSIVE (Multiple Agent Simulation System in Virtual Environment) is a high-end computer animation and artificial intelligence software package used for generating crowd-related visual effects for film and television. It was developed to create fighting armies of hundreds of thousands of soldiers for Peter Jackson's The Lord of the Rings films automatically.[17] https://www.massivesoftware.com/massive101.html Uses `fuzzy logic`.

https://www.youtube.com/watch?v=042YjQ9aZNk

https://www.youtube.com/watch?v=8jd2bWfqN0U

https://www.youtube.com/watch?v=cr5Cwz-5Wsw&t=33s

5.5 Social Media Algorithms

Social media platforms do not show you content in the order it was posted. They use AI systems to construct a personalized feed — a sequence of posts, videos, images, and advertisements assembled to maximize the time you spend on the platform and the actions you take while there.

These systems analyze hundreds of signals: what you have liked, shared, commented on, or paused to read; who you follow and how you interact with them; what content similar users engage with; how long you spend looking at different types of posts; and what you have searched for or clicked on elsewhere on the internet.

The goal, in the language of the industry, is engagement. In practice, this means the algorithm rewards content that produces strong emotional reactions — content that makes you angry, anxious, excited, or amused enough to interact with it. Content that is calm, nuanced, or complex tends to generate less interaction and is shown to fewer people.

Research on the effects of social media algorithms is ongoing, complex, and contested. Several findings recur across studies:

- Emotional contagion — negative and outrage-inducing content spreads further and faster than positive or neutral content
- Radicalization pathways — recommendation systems can lead users from mainstream content to increasingly extreme material through a series of small steps, each appearing to be a minor escalation
- Misinformation spread — false information that provokes strong emotional reactions often outperforms accurate information in algorithmic ranking
- Mental health impacts — particularly among adolescents, high social media use is associated with elevated rates of anxiety, depression, and body image concerns, though the causal mechanisms are debated

Social media companies have access to extensive internal research on the effects of their algorithms. In several cases, this research has shown significant harms — particularly to younger users — that the companies did not disclose publicly. The gap between what these companies know about the effects of their systems and what they choose to communicate publicly is a serious ethical concern. The difficulty of regulation is that algorithmic systems are proprietary, complex, and constantly changing. External researchers typically cannot access the data needed to study their effects. This opacity makes accountability difficult and places enormous trust in companies whose primary obligation is to shareholders rather than to the public.

5.6 AI In Journalism and News

Journalism was among the first creative industries to adopt AI tools at scale, and some applications are straightforwardly beneficial. Automated reporting tools have been used for years to generate routine factual reports from structured data. Earnings reports, sports results, weather summaries, and election results can all be generated automatically from data feeds in seconds, allowing journalists to focus on analysis and investigation rather than transcription.

AI tools also assist with:

- Data journalism — analyzing large datasets to identify patterns, anomalies, or stories that would take humans much longer to find
- Translation — making journalism accessible across language barriers quickly and cheaply
- Transcription — converting audio recordings of interviews and press conferences into searchable text
- Fact-checking assistance — cross-referencing claims against databases of verified information
- Archive search — surfacing relevant historical reporting quickly when a news event breaks

The same capabilities that make AI useful in journalism also create serious risks.

Misinformation generation: AI language models can generate plausible-sounding news articles about events that never happened, complete with fabricated quotes, statistics, and context. These can be produced at scale and distributed across social media before human fact-checkers can respond.

Source fabrication: AI tools have been shown to generate convincing citations, quotes, and expert sources that do not exist. Journalists and editors who rely on AI tools without careful verification risk publishing false information in good faith.

Job displacement: Several major news organizations have laid off staff journalists and replaced their work with AI-generated content. The quality of local and specialist journalism — which depends on human relationships, expertise, and accountability — is threatened when the economic model no longer requires human reporters.

Personalized news and fragmentation: When AI curates each person's news feed based on their preferences, the result can be a fragmented information landscape in which people in the same community have access to entirely different sets of facts. Shared factual reality is a prerequisite for democratic deliberation; algorithmic personalization works against it.

Journalism serves a critical function in democratic societies: it provides citizens with the information they need to make decisions, hold power accountable, and participate in public

life. When AI is used to generate fake news, suppress real news, or fragment the shared information environment, it threatens this function directly.

The Catholic tradition has always valued truth as a foundational good — not merely as a preference, but as something to which we have obligations. This is not simply about avoiding individual lies. It is about maintaining the conditions of public trust on which communities depend.

6. A Catholic Perspective

The Eighth Commandment — you shall not bear false witness against your neighbor — is not only about lying in court. It extends to all forms of deception: fraud, calumny, detraction, and the manipulation of public opinion through false representation. In a media environment shaped by AI, this commandment has new and urgent applications.

Deepfakes that fabricate statements by real people, AI-generated news articles about events that never occurred, recommendation systems that amplify misinformation because it drives engagement — all of these violate the spirit of the Eighth Commandment. Catholics are called to be witnesses to truth, not merely passive consumers of whatever information algorithms place in front of them.

Catholic teaching affirms that human beings are made in the image of God — imago Dei — and that this dignity extends to our creative capacities. Art, music, storytelling, and journalism are not merely economic activities. They are expressions of the human person's participation in God's ongoing creation. They require vulnerability, judgement, moral seriousness, and lived experience that no statistical model can replicate.

When AI is used to replace human creative work rather than to assist it, or when artists' work is used without consent to train systems that then compete with them, it is not merely an economic problem. It is a question of whether we value the human creativity that these systems are built upon.

Catholic social teaching holds that the goods of creation are meant for all, and that concentrated power over common goods requires special justification and accountability. A small number of technology platforms now exercise extraordinary influence over what information billions of people receive, what music they hear, and what stories they are told. This concentration of power is not neutral. It shapes culture, opinion, politics, and belief at scale. The companies exercising this power are accountable primarily to shareholders, not to the communities they affect. Catholic social thought — rooted in subsidiarity, solidarity, and the common good — calls for structures of accountability that ensure technology serves people rather than the other way around.

While structural responses — regulation, platform accountability, collective action — are necessary, they are not sufficient. Personal virtue also has a role. The tradition of prudence —

practical wisdom about how to act well in particular circumstances — applies directly to how we use media.

Prudence in a media environment shaped by AI might mean:

- Seeking out information actively rather than consuming only what algorithms provide
- Pausing before sharing content that provokes strong emotional reactions, especially if it cannot be verified
- Being honest about the source of creative work, whether AI-assisted or not
- Recognizing the difference between entertainment and truth, and not allowing the former to crowd out the latter
- Attending to the effects of media consumption on one's own mind, heart, and relationships

7. Exit Ticket

AI in entertainment and media is not a single technology but a collection of systems with different purposes, different capabilities, and different risks. What they share is the capacity to shape what we see, hear, and believe at a scale and speed that no previous technology has matched.

8. Lesson Summary

This lesson examined six major applications of AI in entertainment and media, and the Catholic ethical principles most relevant to evaluating them.

Section	Highlights
Streaming recommendations	Collaborative filtering, filter bubbles, what "engagement" really optimizes for
AI-generated music & art	How generative models work, training data consent issues, the royalties question
Deepfakes	GANs explained, four distinct risk categories, detection arms race
Film & TV production	De-ageing, script analysis AI, the 2023 writers' strike and what it revealed
Social media algorithms	How feeds are built, emotional contagion, radicalization pathways, platform opacity
AI in journalism	Automated reporting benefits vs. misinformation generation, source fabrication, news fragmentation
Catholic perspective	Eighth Commandment, *imago Dei*, common good and platform power, prudence as personal virtue

Key Takeaways

- Recommendation systems are not neutral tools — they are designed to maximize engagement, which is not the same as maximizing your wellbeing or exposure to truth
- AI can generate music, art, and video indistinguishable from human work, raising fundamental questions about creativity, authorship, and economic justice
- Deepfakes pose a direct threat to truth, trust, and human dignity — and detection tools consistently lag behind generation capabilities
- AI is changing how films and TV are made, with significant implications for creative risk-taking and for workers in the industry
- Social media algorithms amplify emotionally provocative content, contributing to polarization, misinformation spread, and documented mental health harms
- AI in journalism offers genuine benefits but also enables misinformation at scale and threatens the economic model for human reporting
- Catholic teaching on truth, human dignity, and the common good provides clear principles for evaluating these technologies, even if the specific applications are new

10. Vocabulary

Term	Definition
State machine / finite state machine (FSM)	A system in which an entity (like an NPC) is always in exactly one defined state — such as patrol, chase, or investigate — and switches between states based on specific triggering events.
Procedural generation	Using algorithms and mathematical rules — rather than human designers — to automatically create game content such as maps, terrain, items, and entire worlds.
AI Director	A meta-level AI system that monitors the entire game session and dynamically adjusts pacing, enemy spawning, and resource placement to maintain emotional tension. Most famously used in Left 4 Dead.
Behavior tree	A hierarchical structure of decisions and actions used to control NPC behavior. More flexible and powerful than a simple state machine, allowing for richer, more nuanced character logic.
Generative AI	AI systems that can produce new content — images, text, audio, video, or code — by learning patterns from large datasets of existing human-created work.
Computer vision	A field of AI that enables computers to interpret and understand visual information from images or video, such as identifying objects, detecting defects, or reading facial expressions.
Recommendation system	An AI system that predicts content a specific user is likely to enjoy or engage with, based on their history and the behavior of similar users. Used by Netflix, Spotify, YouTube, and others.
Generative adversarial network (GAN)	A type of AI model in which two neural networks compete: one generates fake content, the other tries to detect it. Over many cycles this produces highly realistic synthetic outputs, including deepfakes.

Term	Definition
Deepfake	AI-generated audio, video, or image content in which a real person's face, voice, or likeness has been fabricated or replaced. Derived from "deep learning" and "fake."
Voice cloning	Using AI to generate speech in a specific person's voice from a short audio sample, producing new recordings of things they never said.
Misinformation / disinformation	Misinformation is false content spread without intent to deceive. Disinformation is false content spread deliberately to mislead. AI can generate and amplify both at scale.
Personalization	The use of AI to tailor digital experiences — content, ads, news, recommendations — to individual users based on their data and behavior.
Imago Dei	Latin for "image of God." The Catholic teaching that every human being is created in God's image and therefore possesses inherent, inalienable dignity that no technology can replicate or replace.
Common good	In Catholic social teaching, the set of social conditions that allow individuals and communities to flourish fully. Technology should serve the common good rather than concentrate power or undermine public trust.
Subsidiarity	A Catholic social principle holding that decisions should be made at the most local level possible, and that larger institutions should not take over what smaller communities can handle. Applied to technology, it cautions against excessive concentration of power in a few platforms.

9. Next Week!

Next week we will explore AI's impact on employment. We will discuss the types of jobs AI can replace (factory work, data entry) versus those it will enhance (creative fields, engineering). Students will analyze reports on the future job market and predict which careers AI might create. The class ends with a debate on whether AI will create more jobs than it eliminates.